Nifty futures: factors that affect it, and their backtest (train / valid / test, full and regime-wise)

Author: Ankit Garg. Research only, not investment advice. **All numbers use real NSE Nifty futures prices** (daily bhavcopy, front month, premium and roll included). No spot-index results are used; the earlier spot 18% result is dropped.

Format of every cell: **CAGR / MaxDrawdown / Sharpe**. Periods: Train 2017-04 to 2021-12, Valid 2022-23, Test 2024-01 to 2026-10. Signal decided at 15:00 using only earlier data; fill at the last-30-minute VWAP; cost 0.025% per side plus 0.05% per roll (see section 3 for realistic costs).

Two views of return: **pure futures P&L** (price change including the premium decay a long pays; no interest) and **total return** (pure P&L plus 6% p.a. on capital, since futures need only margin). The pure view is the honest answer to the carry question; the total view is how idle cash is normally treated.

1. How many strategies were tried (overfitting accounting)

Search	Futures variants
Real-futures search `rf_search.py` (signals x sizing x mode)	378
Factor permutation search `rf_factor_perm.py` (6 bases x 0-3 of 20 gates x 2 sizings x 2 modes)	32,424
Earlier spot-index searches (about 800 gates, not futures; excluded from all numbers)	not counted
Total futures variants looked at	**about 32,800**

Picks were made on **train + valid only** (score = the weaker of the two periods' CAGR minus |MaxDD|). Test was never used to choose or change anything. After the picks, ablation and neighbour-threshold runs were done as diagnostics only; they did not change the strategy.

Because ~32,800 variants were tried, the best train+valid score is optimistic. Check on test: of the 100 best train+valid variants, the median test CAGR was 3.8% (buy-and-hold futures 0.4%); 87% beat buy-and-hold on test CAGR and 95% had a smaller test drawdown. The single top variant ranks only at the 40th percentile of all variants on test (CAGR minus drawdown). Rank correlation between train+valid and test score is 0.28: real but weak.

2. Frozen strategies vs simply holding Nifty futures

- **Hold** = buy Nifty futures and keep them (rolling each month).
- **A: Trend(50) + rising OI**: long when the 15:00 price is above its 50-day average AND total futures open interest rose over the last 5 days (OI known to the previous close). Otherwise flat.
- **B: Factor-gated**: long unless USD/INR rose >2% in 21 days, OR US 10Y yield rose >=25bp in 21 days, OR it is January/February. Chosen as best of the 32,424 on train+valid.

Total return (futures P&L + 6% interest on capital):

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
Hold futures	16.7% / -38.3% / 0.93	11.9% / -15.3% / 0.88	0.4% / -18.6% / 0.10	10.6% / -38.3% / 0.71
A: Trend50 & OI	11.3% / -8.2% / 1.47	12.8% / -2.9% / 2.25	4.2% / -7.1% / 0.70	9.4% / -8.2% / 1.38

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
B: Factor-gated	25.7% / -10.1% / 2.04	23.9% / -5.2% / 2.57	8.1% / -10.0% / 0.77	19.8% / -10.1% / 1.73

Pure futures P&L (no interest; includes the premium a long pays):

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
Hold futures	9.9% / -41.0% / 0.60	5.4% / -17.4% / 0.45	-5.4% / -27.9% / -0.35	4.2% / -41.0% / 0.34
A: Trend50 & OI	4.8% / -10.9% / 0.66	6.2% / -3.8% / 1.15	-1.8% / -8.6% / -0.27	3.1% / -10.9% / 0.48
B: Factor-gated	18.4% / -11.0% / 1.52	16.6% / -5.5% / 1.86	1.8% / -14.9% / 0.22	12.8% / -14.9% / 1.18

Reading: in total-return terms both beat holding futures on Sharpe and drawdown in every period (A's CAGR is lower than holding in train, 11.3% vs 16.7%, and higher in valid and test). In pure-price terms holding futures lost money in test (-5.4%), A is slightly negative (-1.8%) and B slightly positive (+1.8%). A cuts drawdown by roughly 60-80% but does not make high returns; B earns more but with a bigger drawdown than A. Neither makes real money from price alone in the test years.

3. Costs: brokerage, STT, exchange charges, slippage

Assumed per futures lot of about Rs 15 lakh notional: brokerage ~Rs 20/order (~0.0013%), STT 0.02% on the sell side (a 0.05% STT is covered by the x1.6 scenario below), exchange charges 0.00183%, SEBI 0.0001%, stamp duty 0.002% on buy, GST 18% on brokerage and exchange charges, slippage 1bp per side (Nifty futures are very liquid). Round trip comes to about **0.048%**, i.e. 0.024% per side, almost exactly the 0.025% per side the model already used. Scenarios: x1 (0.025%/side), x1.6 (0.04%/side: higher STT plus 2bp slippage), x3 (stress).

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
A costs x1	11.3% / -8.2% / 1.47	12.8% / -2.9% / 2.25	4.2% / -7.1% / 0.70	9.4% / -8.2% / 1.38
B costs x1	25.7% / -10.1% / 2.04	23.9% / -5.2% / 2.57	8.1% / -10.0% / 0.77	19.8% / -10.1% / 1.73
A costs x1.6	10.4% / -8.6% / 1.36	12.1% / -2.9% / 2.14	3.5% / -7.4% / 0.59	8.6% / -8.6% / 1.27
B costs x1.6	25.1% / -10.1% / 2.00	23.2% / -5.2% / 2.51	7.6% / -10.5% / 0.72	19.2% / -10.5% / 1.69
A costs x3	8.3% / -9.6% / 1.11	10.4% / -3.2% / 1.86	1.8% / -7.9% / 0.31	6.8% / -9.6% / 1.01
B costs x3	23.6% / -10.2% / 1.90	21.7% / -5.3% / 2.37	6.4% / -11.7% / 0.62	17.8% / -11.7% / 1.58
Hold futures (1 entry, ~1 roll/month)	16.7% / -38.3% / 0.93	11.9% / -15.3% / 0.88	0.4% / -18.6% / 0.10	10.6% / -38.3% / 0.71

Trading frequency: A holds 35% of days, about 20 round trips a year; B holds 64% of days, about 9 round trips a year. Low turnover is why costs barely matter; at 3x costs both still beat holding futures on Sharpe and drawdown (A's test CAGR falls to 1.8%).

4. Parameter check: is 50 a fluke?

Trend(N) AND OI(5-day) rising, total-return view:

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
Trend(30) & OI5	9.7% / -8.2% / 1.26	15.8% / -2.6% / 2.65	1.8% / -8.9% / 0.34	8.5% / -8.9% / 1.25
Trend(40) & OI5	11.2% / -7.5% / 1.46	13.2% / -3.7% / 2.28	1.5% / -9.9% / 0.27	8.6% / -9.9% / 1.26

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
Trend(50) & OI5	11.3% / -8.2% / 1.47	12.8% / -2.9% / 2.25	4.2% / -7.1% / 0.70	9.4% / -8.2% / 1.38
Trend(60) & OI5	10.2% / -8.2% / 1.32	10.4% / -3.7% / 1.82	5.4% / -7.1% / 0.89	8.8% / -8.2% / 1.28
Trend(80) & OI5	7.1% / -9.6% / 0.96	10.4% / -3.8% / 1.73	5.7% / -6.1% / 0.93	7.4% / -9.6% / 1.08
Trend(100) & OI5	7.5% / -7.7% / 1.01	11.1% / -4.4% / 1.84	5.9% / -6.1% / 0.94	7.8% / -7.7% / 1.13
Trend(150) & OI5	8.8% / -7.0% / 1.18	13.1% / -4.9% / 1.90	7.1% / -6.1% / 1.11	9.2% / -7.0% / 1.30

Same, pure futures P&L:

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
Trend(30) & OI5	3.3% / -10.0% / 0.47	9.0% / -3.6% / 1.58	-4.1% / -11.9% / -0.69	2.2% / -11.9% / 0.36
Trend(40) & OI5	4.7% / -9.3% / 0.65	6.5% / -4.9% / 1.19	-4.5% / -12.6% / -0.72	2.3% / -12.7% / 0.37
Trend(50) & OI5	4.8% / -10.9% / 0.66	6.2% / -3.8% / 1.15	-1.8% / -8.6% / -0.27	3.1% / -10.9% / 0.48
Trend(60) & OI5	3.8% / -10.9% / 0.53	3.9% / -4.9% / 0.73	-0.7% / -8.6% / -0.09	2.4% / -10.9% / 0.39
Trend(80) & OI5	0.9% / -12.3% / 0.16	3.9% / -4.9% / 0.70	-0.5% / -7.6% / -0.04	1.1% / -12.3% / 0.20

Reading: N = 30 to 150 all give a similar shape (full CAGR 7.4-9.4%, drawdown 7-10% in total-return view), so 50 is not a lucky spike. Honest caveats: in pure futures P&L the test period is negative for every N (-0.5% to -4.5%), and the shorter averages (30, 40) are clearly worse in test than the longer ones, so 50 to 150 are the safer range.

5. Expiry effect on OI

How handled: the OI used is the **sum of open interest across all live Nifty contracts**, not the front month. When positions roll from the expiring contract to the next one, OI moves between contracts but the sum barely changes, so the roll does not create a fake rise or fall. Checks:

• Front-month-only OI would be wrong around expiry: it agrees with total-OI on 53% of all days but only 32% on days within 3 days before expiry.

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
A with total OI (used)	11.3% / -8.2% / 1.47	12.8% / -2.9% / 2.25	4.2% / -7.1% / 0.70	9.4% / -8.2% / 1.38
A with front-month OI only	12.3% / -7.0% / 1.63	8.7% / -4.3% / 1.70	-4.2% / -13.6% / -0.65	6.4% / -14.0% / 0.99
A with OI gate frozen in last 3 days before expiry	11.2% / -8.8% / 1.36	14.5% / -2.9% / 2.50	4.2% / -5.6% / 0.71	9.7% / -8.8% / 1.36

Reading: using total OI (or freezing the gate near expiry) gives the same results; the front-month-only version is the one that is distorted (test CAGR -4.2% instead of +4.2%). Rolls also pay 0.05% in the cost model whenever a position is carried across expiry.

6. Timing: no future data in the signal

• Decision is made at 15:00 on day t. The price used is the 15:00 bar close; the position is filled at the last-30-minute VWAP after that.

• OI for day t is published after close, so the signal uses `OI[t-1]` (OI is shifted one day and the 5-day change is taken on the shifted series). The 5-day change at day t is OI(t-1)/OI(t-6)-1.

• All macro series (USD/INR, US10Y, DXY, S&P, Brent, VIX) are the latest close **strictly before** day t; FII monthly flows are used only from the next month.

Proof by experiment:

Strategy	Train 2017-21	Valid 2022-23	Test 2024-26	Full
A as used (OI lagged 1 day)	11.3% / -8.2% / 1.47	12.8% / -2.9% / 2.25	4.2% / -7.1% / 0.70	9.4% / -8.2% / 1.38
A cheating: same-day OI (not allowed)	6.7% / -12.7% / 0.82	10.4% / -4.4% / 1.98	4.6% / -6.1% / 0.74	6.8% / -12.7% / 0.95
A with OI lagged 2 days	11.7% / -7.3% / 1.48	11.7% / -3.1% / 1.98	3.8% / -8.1% / 0.53	9.3% / -8.1% / 1.26

A leak would make the used version look better than the safer lagged versions. It does not: lag-1 (9.4% full) is about the same as lag-2 (9.3%), and the same-day OI row is not better (6.8%). So the one-day lag is not hiding any advantage, and no future OI is used. The OI series and gates are built from the previous close only.

7. Factors from the Nifty document: what each one does (standalone, on futures)

Each factor is a gate on 'always long futures': allow holding only when the factor is calm. 'Blocked-day return' is the annualised pure-futures return on the days the gate said stay out. **A good factor has a blocked-day return below the allowed-day return.** Train / Valid / Test shown separately.

Factor (caution when…)	Days blocked	Blocked-day ann. return Train / Valid / Test	Allowed-day ann. return Train / Valid / Test	Gated-long (total return) Train / Valid / Test CAGR	Verdict
Rupee weak: USD/INR +2% in 21d	11%	-31% / 97% / 19%	19% / -1% / -7%	22.4% / 5.6% / -1.0%	mixed
Rupee weak: +0.3% yesterday	17%	23% / -1% / -23%	7% / 6% / -3%	10.0% / 10.4% / 2.5%	works in 2 of 3
Dollar index +1.5% in 21d	20%	20% / 16% / -27%	9% / -1% / 1%	13.4% / 5.6% / 6.4%	mixed
US 10Y +25bp in 21d	15%	-36% / -29% / -20%	14% / 32% / -3%	19.8% / 26.1% / 3.2%	works in all 3
S&P 500 down >2% in 21d	18%	-7% / -10% / 51%	13% / 14% / -12%	17.6% / 15.7% / -4.8%	works in 2 of 3
S&P 500 below 200DMA	21%	-7% / 6% / 73%	14% / 4% / -10%	18.0% / 8.8% / -3.3%	mixed
Brent 21d avg >$100	8%	n/a / 19% / 4%	10% / 1% / -7%	16.7% / 7.8% / 0.0%	too few blocked days in some period
Brent +5% in 5d	14%	48% / -14% / -8%	5% / 9% / -5%	10.3% / 12.7% / 1.0%	works in 2 of 3
India VIX >=20	18%	23% / 29% / 53%	6% / -0% / -8%	10.5% / 6.6% / -2.2%	does not work
India VIX >=25	6%	54% / n/a / n/a	6% / 4% / -9%	12.1% / 11.4% / -3.0%	too few blocked days in some period
India VIX +8% yesterday	6%	-57% / -53% / 8%	18% / 9% / -6%	22.5% / 15.1% / -0.7%	works in 2 of 3
FII sold >10k cr last month	47%	8% / 1% / -10%	11% / 10% / 5%	14.2% / 10.5% / 7.7%	works in all 3
FII sold >2 lakh cr in 12m	32%	n/a / -0% / -10%	10% / 11% / 5%	16.7% / 12.1% / 7.7%	too few blocked days in some period
Nifty P/E >=25	42%	12% / n/a / n/a	2% / 6% / -6%	6.4% / 12.8% / 0.4%	too few blocked days in some period
Jan/Feb seasonality	16%	-10% / -31% / -25%	14% / 14% / -1%	18.5% / 18.9% / 5.7%	works in all 3
Korean won weak 2% in 21d	20%	34% / 7% / -27%	7% / 4% / 1%	11.6% / 9.4% / 6.7%	mixed
Bank Nifty below 50DMA	39%	2% / 7% / -3%	15% / 3% / -8%	15.6% / 7.5% / 1.0%	mixed
Futures OI falling over 5d	43%	16% / -11% / -7%	6% / 19% / -4%	8.4% / 16.0% / 2.5%	works in 2 of 3

Factor (caution when…)	Days blocked	Blocked-day ann. return Train / Valid / Test	Allowed-day ann. return Train / Valid / Test	Gated-long (total return) Train / Valid / Test CAGR	Verdict
Futures premium >=5% annualised	49%	-7% / 2% / -20%	21% / 8% / 31%	17.8% / 9.4% / 14.1%	works in all 3

Blocked-day returns are noisy where blocked days are few (Brent >$100, VIX >=25, P/E, FII 12m are 'n/a' where a period has under 20 blocked days; the rupee 2% figure in valid is a handful of crisis days). Gated-long CAGR is in the total-return view. 'Works' means blocked-day return < allowed-day return in that period.

Hold futures (total return) CAGR for reference: 16.7% / 11.9% / 0.4%. Factors the document called 'no effect' (gold, India 10Y yield, yen, US inflation, expiry week) were not tested as drivers.

FII monthly flows: data file ends 27 Jan 2026, so FII gates carry the last value forward after that (stale). P/E file ends 1 Jul 2026. Treat both as less reliable for 2026.

8. Regime-wise results in Train / Valid / Test / Full

Annualised total return % / Sharpe in each market regime (regime known at the decision day; return is the next holding period). Hold = buy-and-hold futures, A = Trend50 & OI, B = factor-gated. n = days. '-' = fewer than 20 days. Where a strategy is mostly in cash, its volatility is tiny and the Sharpe is inflated (values like 25 are artefacts; read the return column).

Trend (price vs 200DMA)

Regime	Period	n	Hold	A	B
Bear (below)	train	205	20.6% / 0.71	17.6% / 2.38	59.3% / 2.80
Bear (below)	valid	105	60.0% / 2.60	7.9% / 2.48	31.1% / 2.05
Bear (below)	test	250	3.3% / 0.28	2.5% / 0.79	10.6% / 0.88
Bear (below)	full	560	18.7% / 0.81	8.8% / 1.76	30.5% / 1.90
Bull (above)	train	768	13.7% / 0.99	12.1% / 1.44	20.9% / 1.84
Bull (above)	valid	386	0.8% / 0.13	14.5% / 2.28	21.4% / 3.00
Bull (above)	test	451	-1.2% / -0.03	5.3% / 0.73	6.9% / 0.70
Bull (above)	full	1607	6.5% / 0.54	10.7% / 1.39	17.2% / 1.68

Volatility (India VIX)

Regime	Period	n	Hold	A	B
High >=22	train	205	43.5% / 1.21	14.1% / 1.46	47.5% / 2.41
High >=22	valid	48	30.7% / 1.24	6.3% / 27.12	54.1% / 2.75
High >=22	test	20	-30.3% / -0.82	25.8% / 1.92	-22.4% / -0.67
High >=22	full	273	33.9% / 1.04	13.5% / 1.49	41.8% / 2.02
Low <15	train	485	7.7% / 0.79	8.2% / 1.26	7.5% / 0.90
Low <15	valid	249	16.6% / 1.65	15.0% / 2.67	27.5% / 3.55
Low <15	test	491	-9.8% / -0.86	3.7% / 0.64	0.1% / 0.05

Regime	Period	n	Hold	A	B
Low <15	full	1225	2.0% / 0.24	7.7% / 1.26	8.2% / 1.00
Mid 15-22	train	481	16.6% / 1.06	13.8% / 1.67	38.9% / 2.81
Mid 15-22	valid	194	0.7% / 0.12	12.2% / 1.84	11.9% / 1.51
Mid 15-22	test	190	37.5% / 2.23	3.9% / 0.64	37.2% / 2.51
Mid 15-22	full	867	17.5% / 1.12	11.2% / 1.50	32.5% / 2.53

Momentum (20d return)

Regime	Period	n	Hold	A	B
Down <-3%	train	143	-33.4% / -0.86	6.4% / 25.27	14.6% / 1.20
Down <-3%	valid	85	57.0% / 2.49	4.9% / 1.64	31.1% / 2.32
Down <-3%	test	112	-20.9% / -1.24	6.6% / 25.67	-2.9% / -0.19
Down <-3%	full	340	-12.7% / -0.33	6.1% / 3.97	12.2% / 1.03
Flat	train	595	24.0% / 1.56	10.4% / 1.33	33.0% / 2.55
Flat	valid	244	-13.1% / -0.98	7.4% / 1.87	13.1% / 1.85
Flat	test	462	9.5% / 0.83	6.4% / 0.92	16.6% / 1.63
Flat	full	1302	11.3% / 0.87	8.4% / 1.21	23.5% / 2.14
Up >3%	train	414	30.7% / 1.99	15.3% / 1.62	21.4% / 1.64
Up >3%	valid	162	34.8% / 2.60	27.0% / 3.02	36.3% / 3.59
Up >3%	test	127	-9.6% / -0.58	-4.8% / -0.73	-9.2% / -0.58
Up >3%	full	704	23.1% / 1.58	13.9% / 1.59	18.3% / 1.45

Drawdown (from 252d high)

Regime	Period	n	Hold	A	B
Deep >10%	train	135	86.7% / 1.72	13.8% / 1.57	109.7% / 3.73
Deep >10%	valid	62	34.9% / 1.63	15.3% / 4.67	4.5% / 0.36
Deep >10%	test	136	15.9% / 0.98	7.8% / 11.37	37.8% / 2.69
Deep >10%	full	333	44.7% / 1.39	11.6% / 2.00	55.2% / 2.73
Near high <5%	train	687	16.6% / 1.29	11.2% / 1.34	22.2% / 2.14
Near high <5%	valid	281	3.1% / 0.34	14.5% / 2.06	19.3% / 2.51
Near high <5%	test	347	-0.6% / 0.01	5.7% / 0.74	3.5% / 0.39
Near high <5%	full	1316	8.9% / 0.76	10.4% / 1.31	16.3% / 1.68
Pullback 5-10%	train	291	-5.1% / -0.25	12.8% / 1.86	9.4% / 0.85
Pullback 5-10%	valid	148	18.6% / 1.11	9.4% / 2.65	41.0% / 5.03
Pullback 5-10%	test	218	-6.7% / -0.45	0.0% / 0.03	0.0% / 0.06

Regime	Period	n	Hold	A	B
Pullback 5-10%	full	658	-0.1% / 0.07	7.6% / 1.35	13.1% / 1.20

Calendar year (full-period strategy returns, total return view)

Year	Hold	A	B
2017	16.5%	0.5%	13.9%
2018	6.4%	15.9%	13.3%
2019	11.8%	14.6%	19.5%
2020	20.5%	16.7%	54.0%
2021	24.6%	6.6%	24.6%
2022	7.0%	10.3%	19.3%
2023	16.9%	15.2%	28.2%
2024	7.3%	7.6%	12.6%
2025	8.5%	5.5%	16.9%
2026	-13.1%	-1.0%	-5.4%

9. Verdict against the checklist

1. Futures only: yes, every number above. Spot results are dropped.

2. Carry: a long pays the premium; it is inside every pure-futures figure. Hold futures made 4.2% a year over the full period in pure terms and lost money in test.

3. Costs: realistic all-in costs are about the 0.025% per side already modelled; at 3x costs both strategies still beat holding futures.

4. Versus holding futures: A and B both beat it on Sharpe and drawdown in train, valid and test (total-return view). In pure futures P&L, B beats holding in every period; A does so only on drawdown and Sharpe, and its price-only return in train (4.8%) is below holding (9.9%).

5. Train / valid / test: shown separately everywhere. Both strategies earn less in test than in train/valid.

6. Expiry: total OI across contracts is used; front-month-only OI is distorted near expiry.

7. Timing: OI lagged one day; lagging two days gives the same result and same-day OI is not better, so there is no hidden look-ahead advantage.

8. Parameters: Trend 30 to 150 behave similarly, not a lucky 50.

9. Overfitting: about 32,800 futures variants tried; picks on train+valid only; test untouched; top-100 median test CAGR 3.8% vs 0.4% for holding.

Honest limits. (a) Test (2024 to Oct 2026) is only 2.8 years, with a flat-to-falling Nifty, so it favours any strategy that sits out. (b) Pure-futures test returns for A and B are about -2% to +2% a year; the headline test numbers rely on interest earned on idle capital. (c) B's thresholds come from a document written with the full 2016-26 history in view, so B is less blind than A. (d) Daily data only; execution at VWAP is assumed.

Suggested next step: paper-trade A (simplest, 2 parameters, smallest drawdown) and B side by side for 2-3 months, with the rules frozen as written here; compare each with simply holding Nifty futures after real costs.